

Chapter 5, Section 3

James Lee

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1. Let X be a nonsingular projective variety of any dimension, let Y be a nonsingular subvariety, and let $\pi : \tilde{X} \rightarrow X$ be obtained by blowing up Y . Show that $p_a(\tilde{X}) = p_a(X)$.

Proof. The proof of (3.4) follows through for the more general case by (II, 8.24).

Proposition 0.1. *In the above setting, we have $\pi_* \mathcal{O}_{\tilde{X}} = \mathcal{O}_X$, and $R^i \pi_* \mathcal{O}_{\tilde{X}} = 0$ for $i > 0$. Therefore, $H^i(X, \mathcal{O}_X) \cong H^i(\tilde{X}, \mathcal{O}_{\tilde{X}})$ for all $i \geq 0$.*

□

3. Let $\pi : \tilde{X} \rightarrow X$ be a monoidal transformation, and let D be a very ample divisor on X . Show that $2\pi^*D - E$ is ample on $\tilde{X}$.

Proof. We use the Nakai-Moishezon Criterion (1.10) to show $D' = 2\pi^*D - E$ is ample. The first provision is easy to check. Since D is ample, we have $D^2 > 0$ in X , so by (3.2), we have

$$D'^2 = 4D^2 - 1 > 0.$$

Let C be any irreducible curve in $\tilde{X}$, and let $r = C.E$. By (3.2), we have

$$D'.C = 2D.(\pi_*C) - r.$$

□

5. Let $a_1, \dots, a_r, r \geq 5$, be distinct elements of k , and let C be the curve in $\mathbb{P}^2$ given by the (affine) equation $y^2 = \prod_{i=1}^r (x - a_i)$. Show that the point P at infinity on the y -axis is a singular point. Compute δ_P and $g(\tilde{Y})$, where $\tilde{Y}$ is the normalization of Y . Show in this way that one obtains hyperelliptic curves of genus $g \geq 2$.

Proof.

□

7. For each of the following singularities at $(0, 0)$ in the plane, give an embedded resolution, compute δ_P , and decide which ones are equivalent.

- (a) $x^3 + y^5 = 0$
- (b) $x^3 + x^4 + y^5 = 0$
- (c) $x^3 + y^4 + y^5 = 0$
- (d) $x^3 + y^5 + y^6 = 0$
- (e) $x^3 + xy^3 + y^5 = 0$

Proof.

- (a) The substitution $x = uy$ gives

$$y^3(u^3 + y^2) = 0,$$

so after the first blow-up, we obtain a cuspidal curve. Thus,

$$E_1 : y = 0, \quad C_1 : u^3 + y^2 = 0,$$

where E_1 is the exceptional locus and C_1 is the strict transform of C . Next, the substitution $y = uv$ gives

$$u^2(u + v^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u + v^2 = 0.$$

The substitution $u = vw$ then gives

$$v(w + v) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + v = 0.$$

Finally, the substitution $v = wt$ gives

$$w(t + 1) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : t + 1 = 0.$$

The total transform of the original curve is

$$w^{15}t^9(t + 1) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}2(2-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) = 4.$$

(b) The substitution $x = uy$ gives

$$y^3(u^3 + u^4y + y^2) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + u^4y + y^2 = 0.$$

Next, the substitution $y = uv$ gives

$$u^2(u + u^3v + v^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u + u^3v + v^2 = 0.$$

The substitution $u = vw$ then gives

$$v(w + v^3w^3 + v) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + v^3w^3 + v = 0.$$

Finally, the substitution $v = wt$ gives

$$w(1 + w^5t^3 + t) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : 1 + w^5t^3 + t = 0.$$

The total transform of the original curve is

$$w^{15}t^9(1 + w^5t^3 + t) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}2(2-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) = 4.$$

(c) The substitution $x = uy$ gives

$$y^3(u^3 + y + y^2) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + y + y^2 = 0.$$

Next, the substitution $y = uv$ gives

$$u(u^2 + v + uv^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u^2 + v + uv^2 = 0.$$

The substitution $v = uw$ then gives

$$u(u + w + u^2w^2) = 0,$$

so the third blow-up gives

$$E_3 : u = 0, \quad C_3 : u + w + u^2w^2 = 0.$$

Finally, the substitution $u = wt$ gives

$$w(t + 1 + w^3t^2) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : t + 1 + w^3t^2 = 0.$$

The total transform of the original curve is

$$w^{12}t^8(t + 1 + w^3t^2) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) = 3.$$

(d) The substitution $x = uy$ gives

$$y^3(u^3 + y^2 + y^3) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + y^2 + y^3 = 0.$$

Next, the substitution $y = uv$ gives

$$u^2(u + v^2 + uv^3) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u + v^2 + uv^3 = 0.$$

The substitution $u = vw$ then gives

$$v(w + v + v^3w) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + v + v^3w = 0.$$

Finally, the substitution $v = wt$ gives

$$w(1 + t + w^3t^3) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : 1 + t + w^3t^3 = 0.$$

The total transform of the original curve is

$$w^{15}t^9(1 + t + w^3t^3) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}2(2-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) = 4.$$

(e) The substitution $x = uy$ gives

$$y^3(u^3 + uy + y^2) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + uy + y^2 = 0.$$

Next, the substitution $y = uv$ gives

$$u^2(u + v + v^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u + v + v^2 = 0.$$

The substitution $u = vw$ then gives

$$v(w + 1 + v) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + 1 + v = 0.$$

The total transform of the original curve is

$$v^9 w^5 (vw)^2 (v^2 w)^3 (w + 1 + v) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}2(2-1) + \frac{1}{2}1(1-1) = 4.$$

The singularities in (a), (b), and (d) are equivalent, while the singularities in (c) and (e) are not equivalent to any of the others. $\square$